

Mount Kenya University



School of Computing and Informatics
BIT3208: ADVANCED WEB DESIGN AND DEVELOPMENT
PROJECT LOGBOOK

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Admission Number: BBIT/2024/54601

Course: Bachelor of Business Information Technology

Unit Code: Advanced Web Design and Development

Lecture Name: Michael Nyoro

Semester : 3.2

Technologies Used: PHP , MySQL, HTML, CSS , Javascript and Xampp

Github Repository: <https://github.com/Isaac-kiarie/Ecommerce-Website-Application.git>

TABLE OF CONTENTS

Week 1

Environment setup

Fig 1: Installation of XAMPP/WAMP/Laragon	3
Fig 2: Apache and MySQL Running	4
Fig 3: Localhost Test Page.....	4
Fig 4: Hello World Test.....	5
Fig 5: Database Connection Test.....	5

Week 2

Wireframes and GUI designs

Fig 1: Hand-drawn Wireframe	6
Fig 2: Dashboard Layout Design.....	7
Fig 3: Mobile View Mockup.....	8
Fig 4: Color Theme and Navigation Structure.....	8
Fig 5: GUI Prototype in Figma/Canva/Draw.io.....	9

Week 3

JavaScript and PHP basics

Fig 1: JavaScript Form Validation.....	10
Fig 2: Password Strength Checker.....	11
Fig 3: PHP Syntax Practice.....	12
Fig 4: Database Connection Script.....	13
Fig 5: Dynamic User Input Handling.....	14

Week 4

Database Design and CRUD Operations

Login form screenshots.....	15
Form validation screenshots.....	16
PHP processing screenshots.....	17
Folder structure screenshots.....	17
Database connection screenshots.....	18
Code snippets.....	18
GitHub updates.....	19

Week 5

Database screenshots.....	20
Table creation screenshots.....	21
CRUD operation screenshots.....	21
Login database integration screenshots.....	21
SQL query screenshots.....	21

GitHub updates.....	22
---------------------	----

Week 6

Tables for the phpmyadmin.....	25
Operations to insert records.....	25
Read operations to display crop records	26
Update operations to edit crop records.....	26
Delete operations.....	27

Week 7

Registration From.....	28
Hashed password.....	29
Login.....	29
Dashboard.....	30
Logout Operation.....	31

WEEK 1: Local Environment Setup

Fig 1.1: XAMPP Installation Success

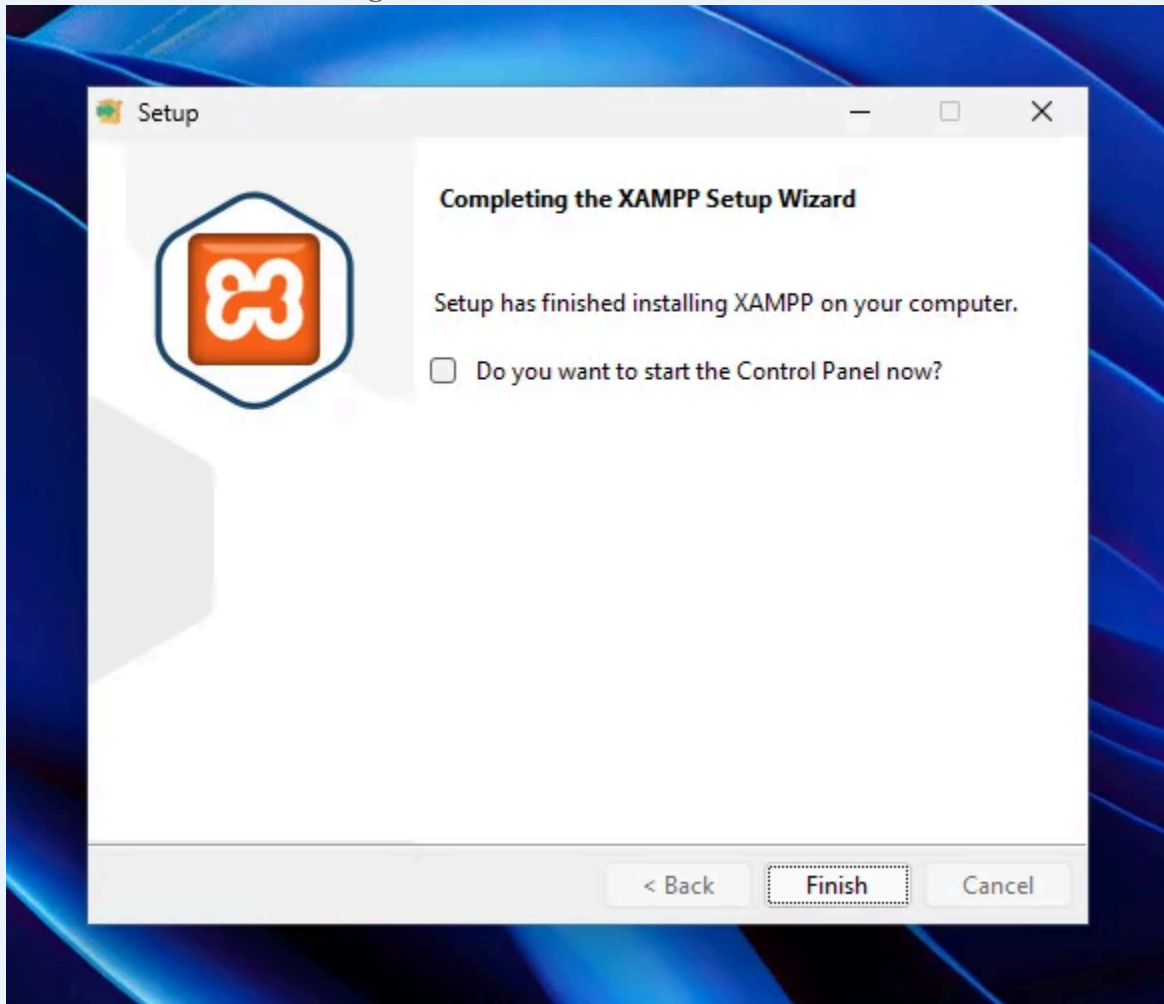


Fig 2: Apache and MySQL Running

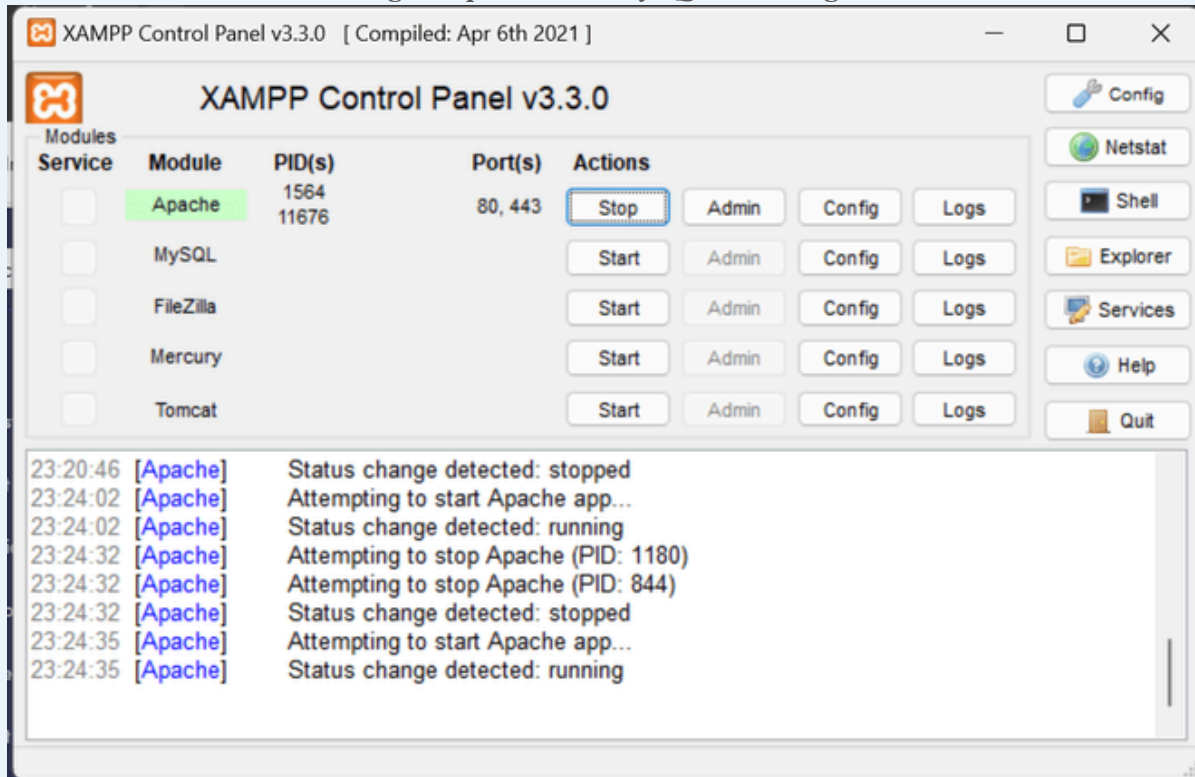


Fig 3: Localhost Test Page

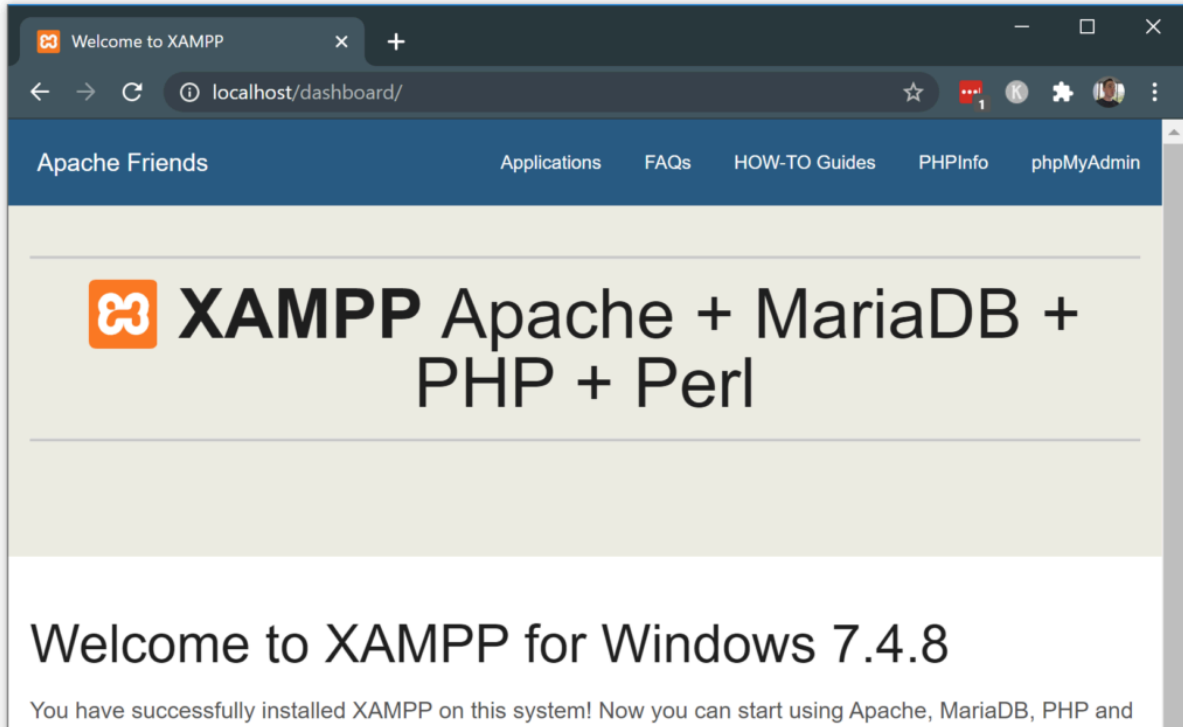
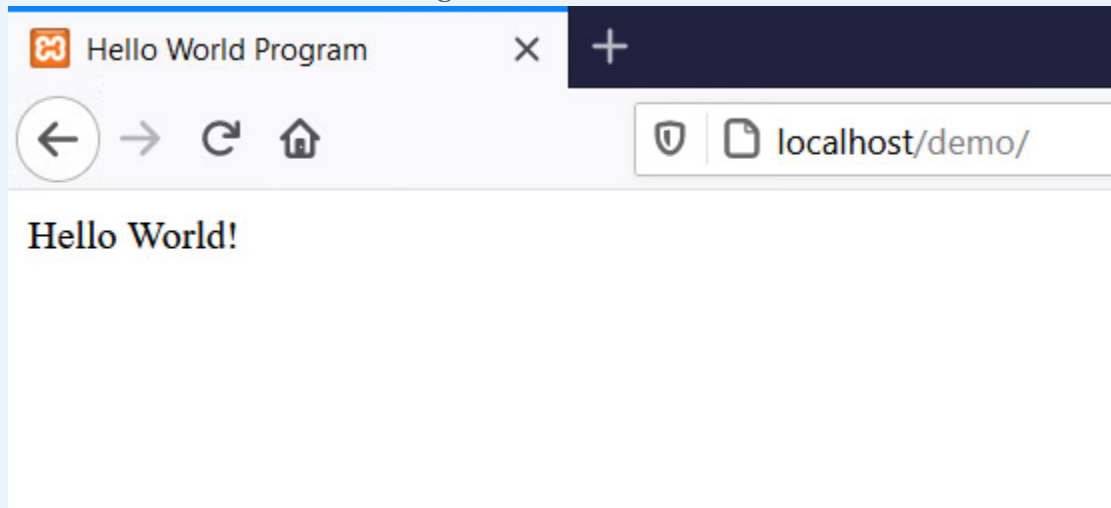


Fig 4: Hello World Test



http://localhost/ecommerce-app/test.php displaying version info and Hello World output.

Fig 5: Database Connection Test



Student Reflection

This week I set up XAMPP on Windows and confirmed that Apache and MySQL services run simultaneously which is essential for a PHP–MySQL application. I also did a hello world test to check if it was functional, the database connection worked well using php and xampp it was easier to connect to it, just a few code it was do

WEEK 2: Wireframe and GUI Design

Fig 1: Hand-drawn Wireframe

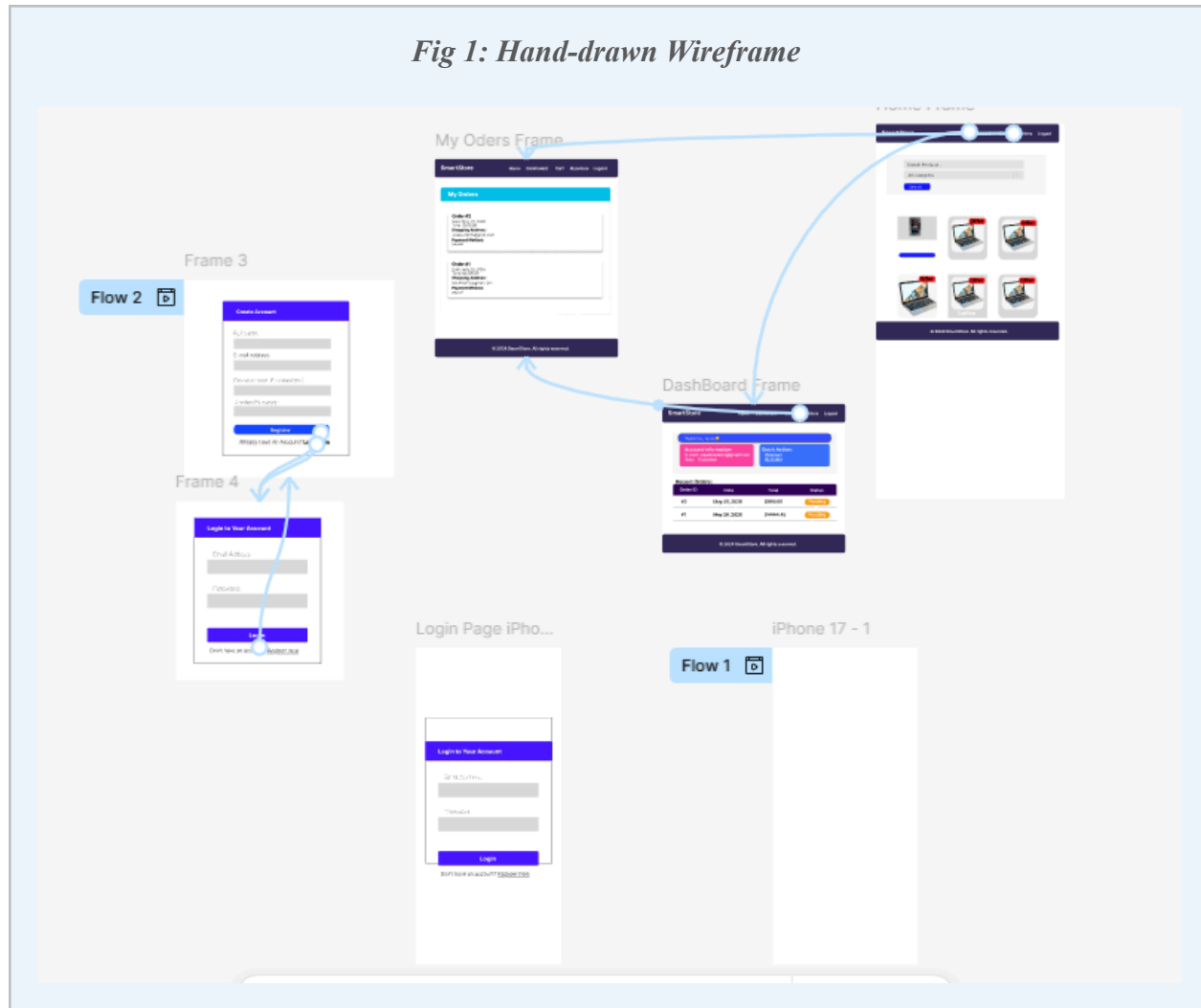


Fig 2: Dashboard Layout Design

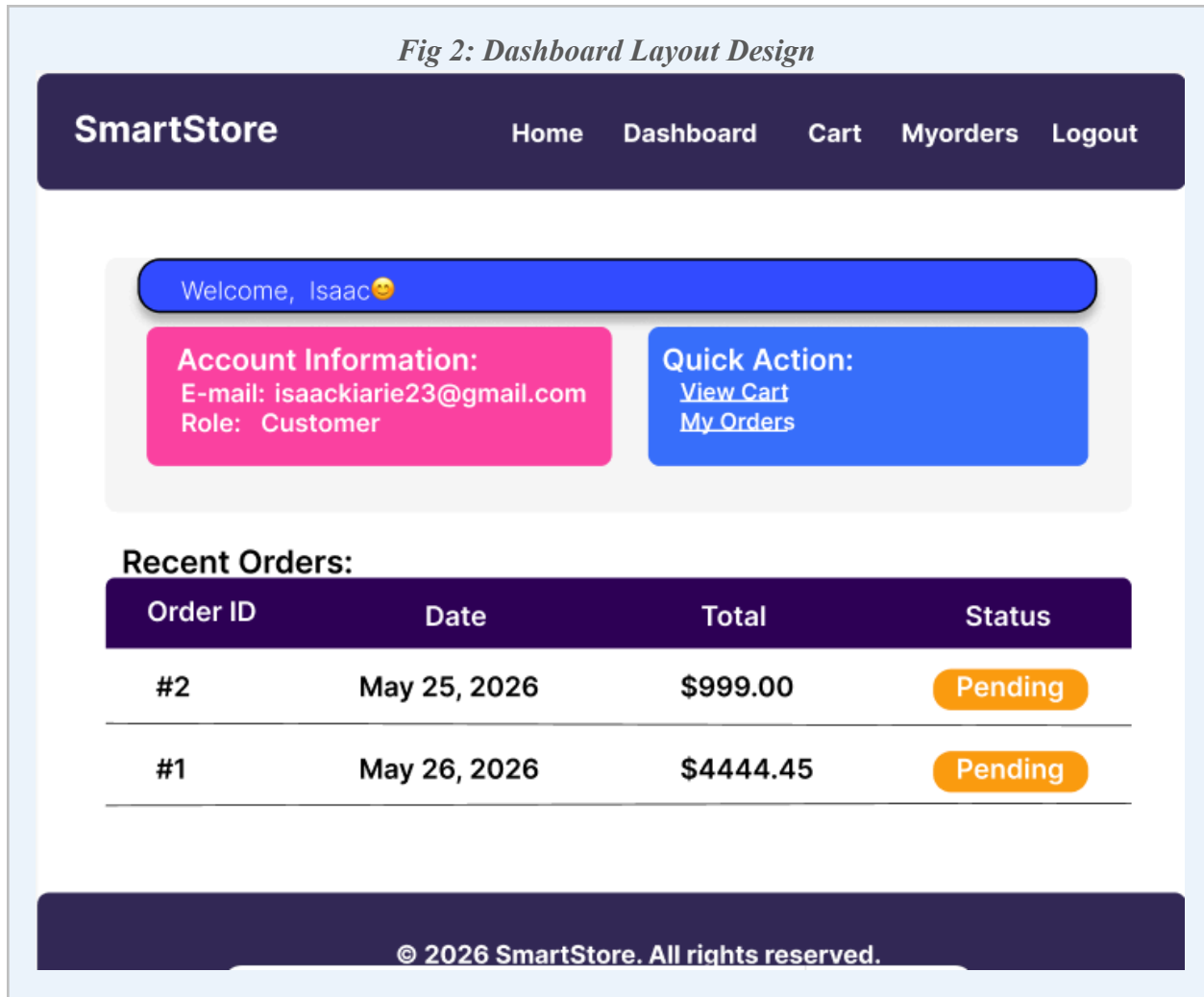


Fig 3: Mobile View Mockup

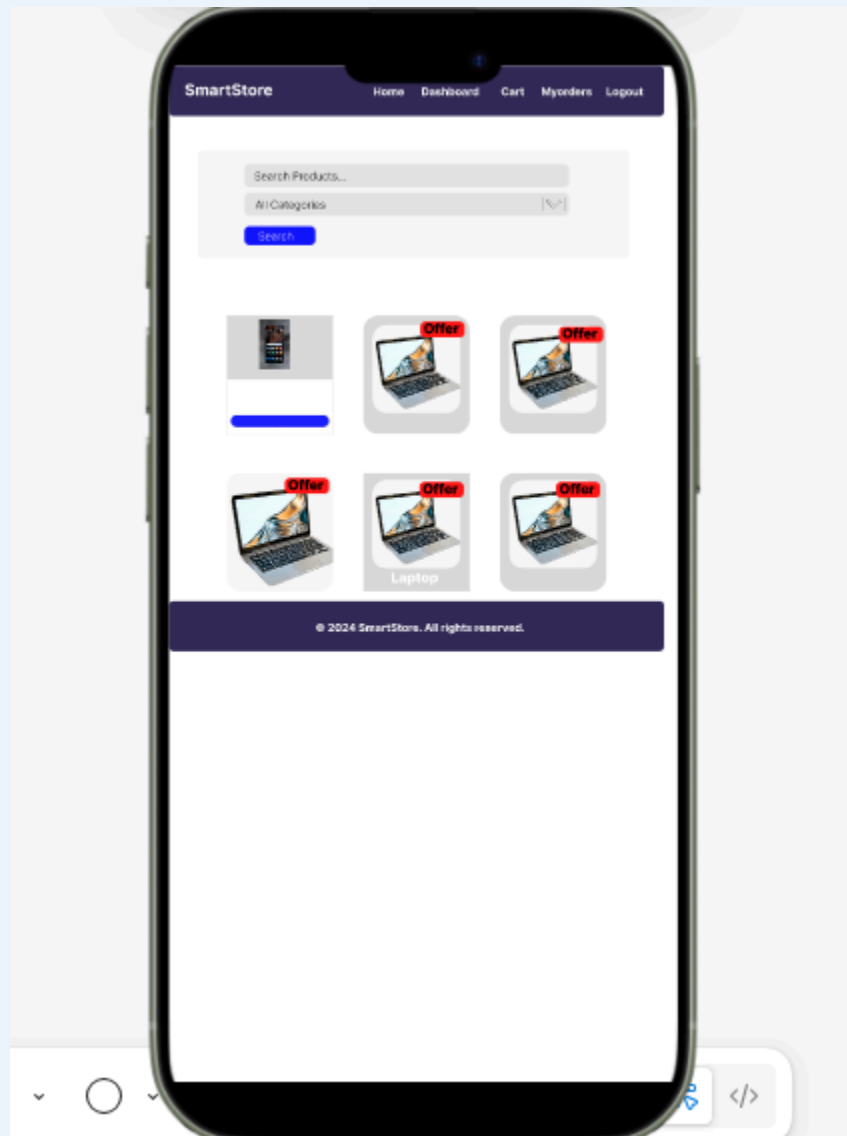
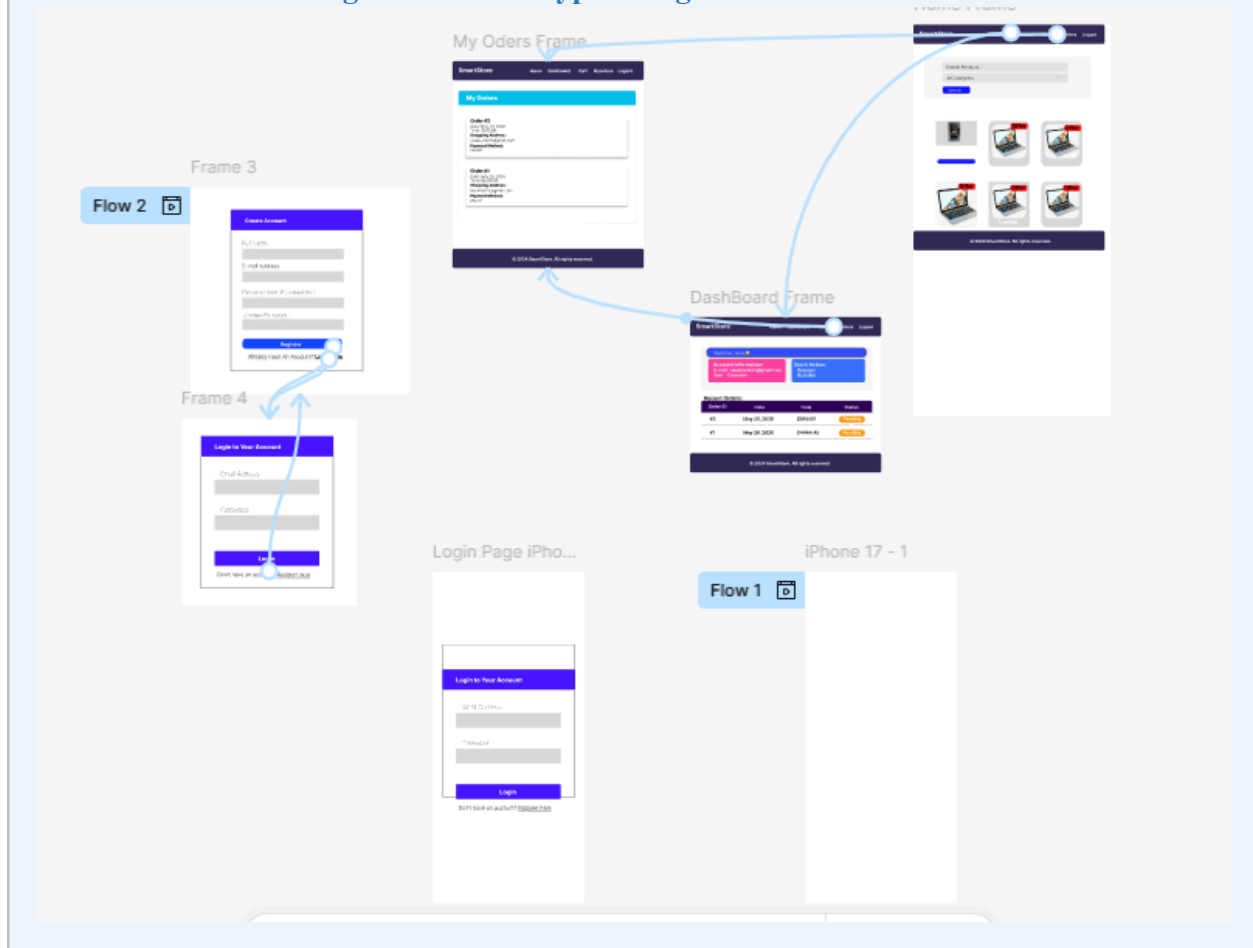


Fig 4: Color Theme and Navigation Structure

SmartStore

Home Login Register

Fig 5: GUI Prototype in Figma/Canva/Draw.io



Student Reflection

Within week two i worked on Hand-drawn Wireframe, Dashboard Layout Design Mobile View Mockup,Color Theme and Navigation Structure GUI Prototype in Figma/Canva/[Draw.io](https://draw.io).

It was awesome and a bit challenging on the mobile mock-up.

I understood Have to use different tools of the figma application used for making the wireframes.

WEEK 3: Javascript and PHP Basics

Fig 1: JavaScript Form Validation

The screenshot shows a web browser window displaying an "EcommerceStore" registration page. The page has a dark blue header with the store name and navigation links for Home, Login, and Register. The main content area features a "Create Account" form with a blue title bar. The form includes a red error message at the top: "Password must be at least 6 characters". Below this, there are four input fields: "Full Name" (containing "John"), "Email Address" (containing "JohnMark12@gmail.com"), "Password (min 6 characters)" (empty), and "Confirm Password" (empty). The browser's taskbar at the bottom shows various application icons and the system clock indicating 09:51.

EcommerceStore Home Login Register

Create Account

Password must be at least 6 characters

Full Name

Email Address

Password (min 6 characters)

Confirm Password

Fig 2: Password Strength Checker

Password

Create a strong password

Password is required

Password Strength: undefined

☐ ✓ At least 8 characters

☐ ✓ At least 1 uppercase letter (A-Z)

☐ ✓ At least 1 lowercase letter (a-z)

☐ ✓ At least 1 number (0-9)

☐ ✓ At least 1 special character (!@#\$%^&*)

☒ ✓ Not a common password

 **Tips for a strong password:**

- Use a mix of uppercase, lowercase, numbers, and symbols
- Make it at least 12 characters for better security

CRUD OPERATION

SmartStoreHomeDashboardCart (2)My OrdersLogout

Shopping Cart

Product	Price	Quantity	Subtotal	Action
T-Shirt Stock: 100 available	\$19.99	1 <button>Update</button>	\$19.99	<button>Remove</button> <button>Save for Later</button>
Novel Stock: 75 available	\$14.99	1 <button>Update</button>	\$14.99	<button>Remove</button> <button>Save for Later</button>

Order Summary
Subtotal (2 items): **\$34.98**
Shipping: Calculated at checkout
Total: **\$34.98**

Proceed to Checkout →Continue Shopping

Fig 3: PHP Syntax Practice

```
1  <?php
2  echo "<!DOCTYPE html>";
3  echo "<html>";
4  echo "<head>";
5  echo "<title>PHP Test</title>";
6  echo "<style>
7      body {
8          font-family: Arial, sans-serif;
9          background: #f4f4f4;
10         text-align: center;
11         padding-top: 100px;
12     }
13     .card {
14         background: white;
15         width: 400px;
16         margin: auto;
17         padding: 30px;
18         border-radius: 10px;
19         box-shadow: 0 0 10px rgba(0,0,0,0.1);
20     }
```

Fig 4: Database Connection Script

```
class Database {
    private $host = "localhost";
    private $db_name = "ecommerce_db";
    private $username = "root";
    private $password = "";
    public $conn;

    public function getConnection() {
        $this->conn = null;

        try {
            $this->conn = new PDO(
                "mysql:host=" . $this->host . ";dbname=" . $this->db_name,
                $this->username,
                $this->password,
                array(
                    PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
                )
            );
        } catch (PDOException $e) {
            echo "Database error: " . $e->getMessage();
        }
    }
}
```

Fig 5: Dynamic User Input Handling

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password



Password is required

Progress bar

Password Strength: undefined

☐ ✓ At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

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Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

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Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

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Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Full name is required

Email Address

Email is required

Password



Password is required

Password Strength: undefined

☐

✓ At least 8 characters

☐

At least 8 characters and 1 number

☐

At least 8 characters, 1 number, and 1 special character

☐

At least 8 characters, 1 number, 1 special character, and 1 uppercase letter

Create Account

Join SmartStore for exclusive deals and secure shopping

Full Name

Enter your full name

Full name is required

Email Address

example@domain.com

Email is required

Password

Create a strong password

Password is required

Password Strength: undefined

At least 8 characters

Student Reflection

Within week 3 i worked on, JavaScript Form Validation, Password Strength Checker, PHP Syntax Practice, Database Connection Script, Dynamic User Input Handling.

My views on this is that I have seen the importance of having a valid credentials from the user and also the ability of the system to indicate to the user the places He/She has gone a miss.

Student Reflection

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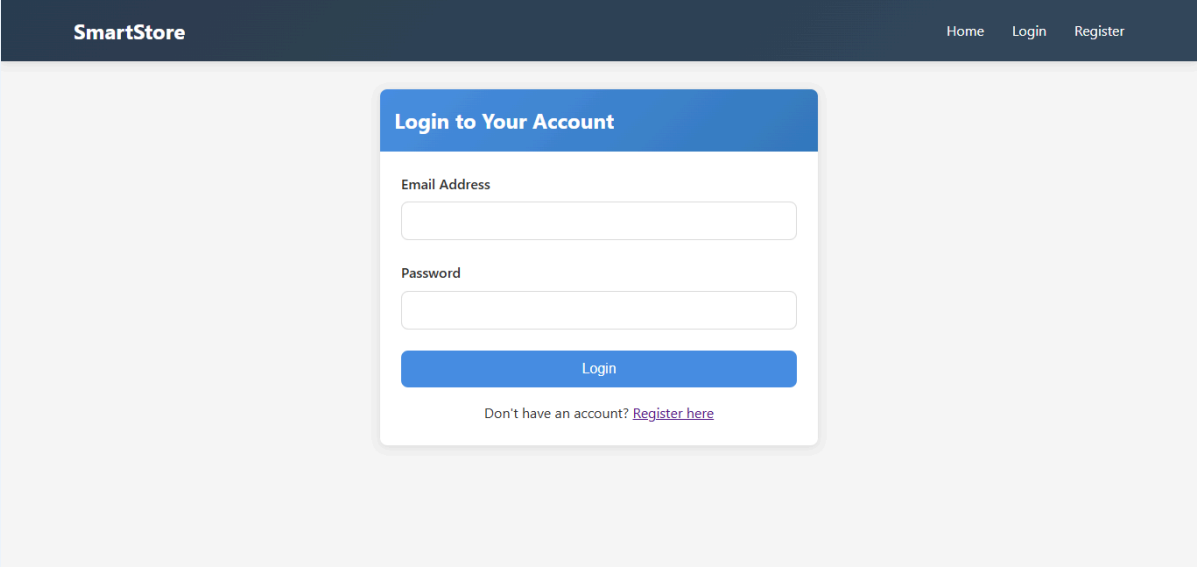
Student Reflection

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My views on this is that I have seen the importance of having a valid credentials from the user and also the ability of the system to indicate to the user the places He/She has gone a miss.

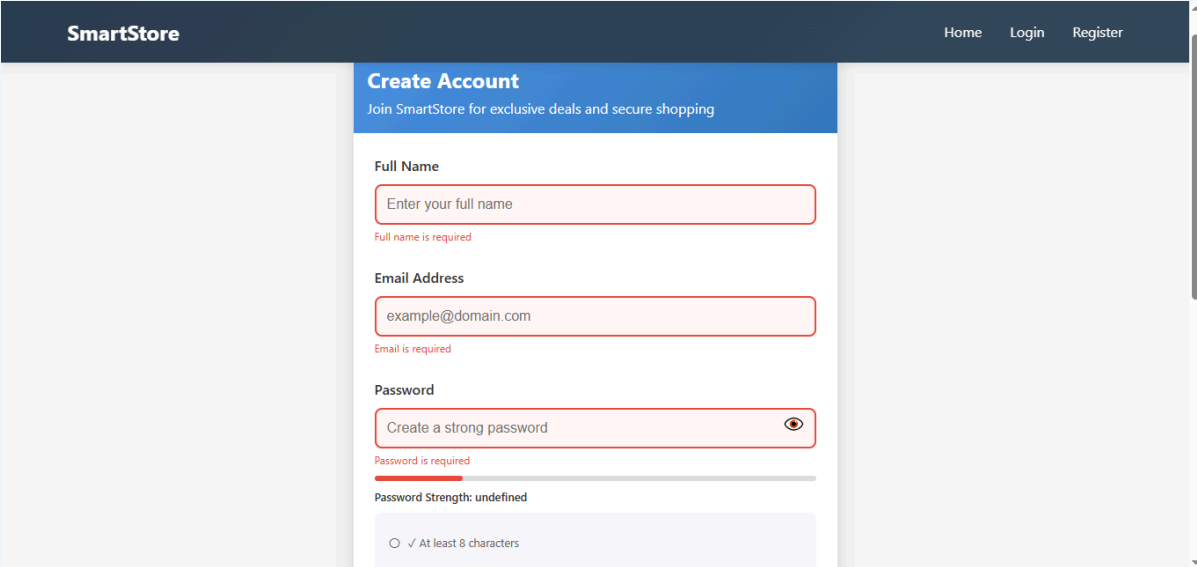
WEEK 4: Database Design and CRUD Operations

Login form screenshots



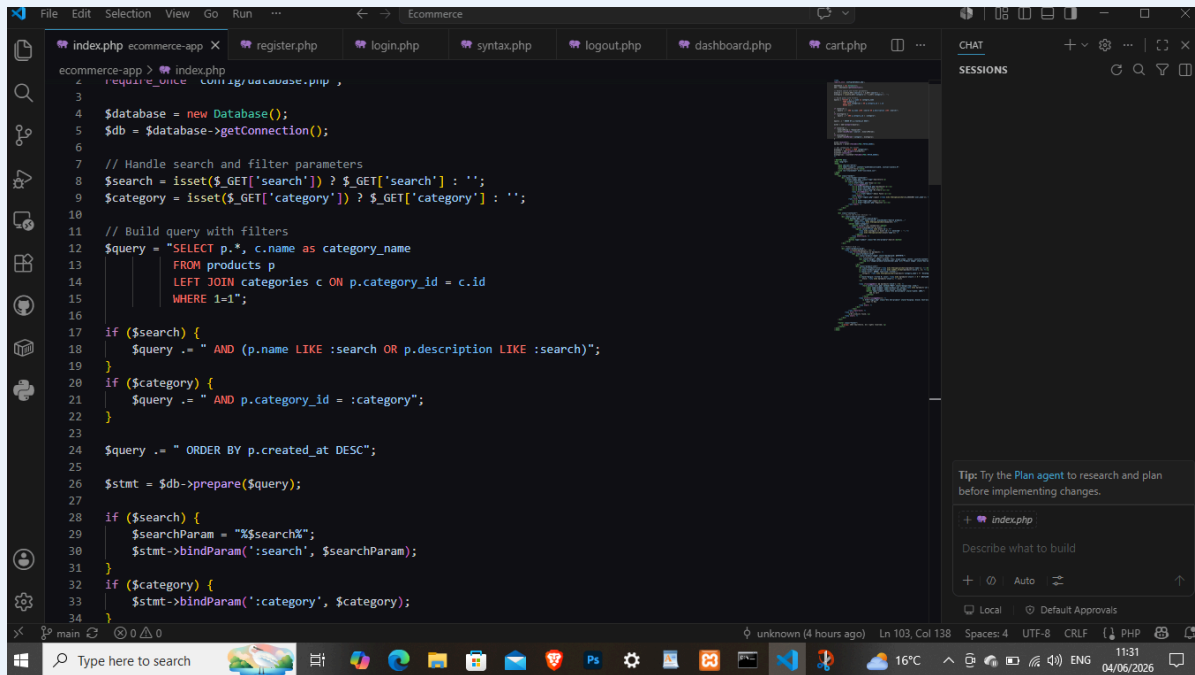
The screenshot shows the 'SmartStore' login page. At the top, a dark blue header contains the 'SmartStore' logo on the left and 'Home', 'Login', and 'Register' links on the right. The main content area is light gray. In the center, there is a white login form with a blue header that says 'Login to Your Account'. The form contains two input fields: 'Email Address' and 'Password'. Below these fields is a blue 'Login' button. At the bottom of the form, there is a link that says 'Don't have an account? [Register here](#)'.

Form validation screenshots



The screenshot shows the 'SmartStore' 'Create Account' page. At the top, a dark blue header contains the 'SmartStore' logo on the left and 'Home', 'Login', and 'Register' links on the right. The main content area is light gray. In the center, there is a white form with a blue header that says 'Create Account' and a sub-header 'Join SmartStore for exclusive deals and secure shopping'. The form contains three input fields: 'Full Name', 'Email Address', and 'Password'. Each field has a red border and a red error message below it: 'Full name is required', 'Email is required', and 'Password is required'. Below the 'Password' field, there is a 'Password Strength: undefined' section with a red progress bar and a radio button labeled 'At least 8 characters'.

PHP Processing screenshots

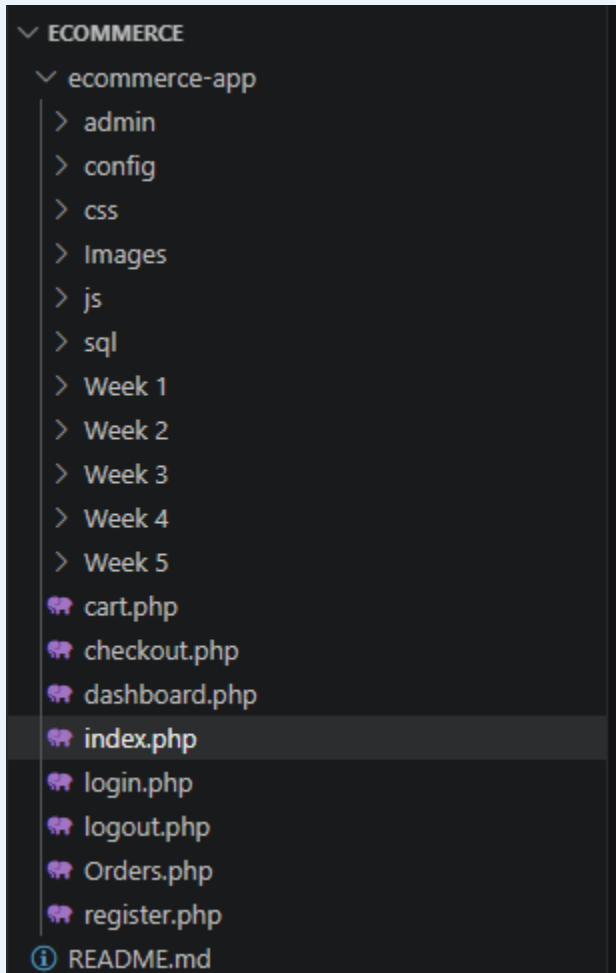


```
File Edit Selection View Go Run ... Ecommerce
index.php ecommerce-app X register.php login.php syntax.php logout.php dashboard.php cart.php ... CHAT
SESSIONS

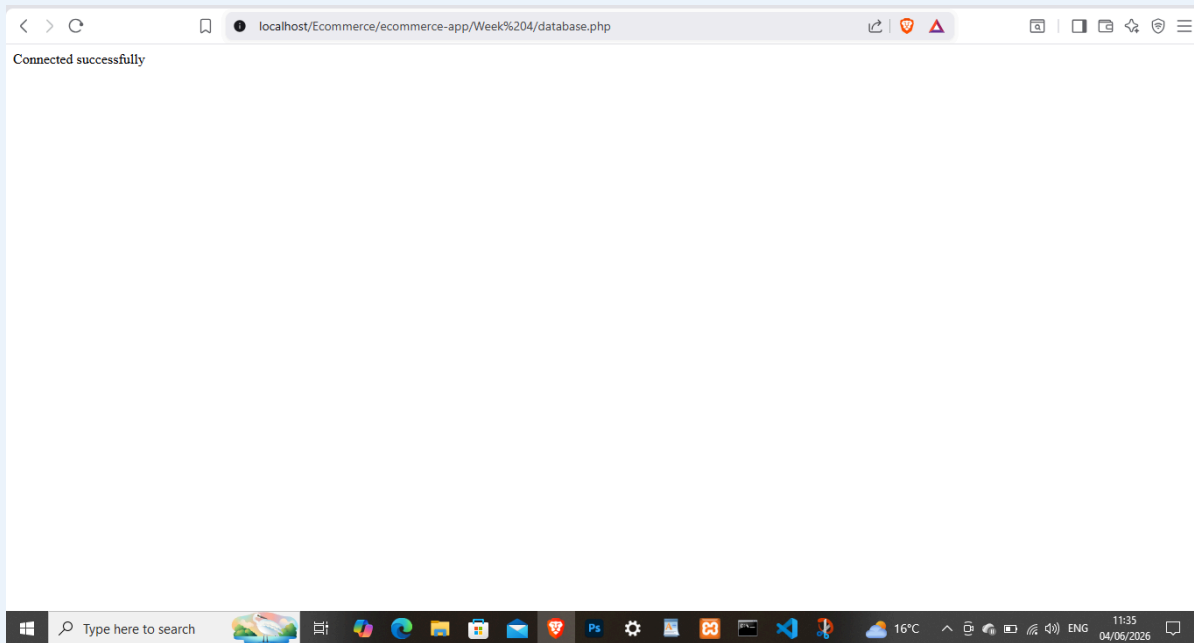
ecommerce-app > index.php
1 require_once 'config/database.php';
2
3
4 $database = new Database();
5 $db = $database->getConnection();
6
7 // Handle search and filter parameters
8 $search = isset($_GET['search']) ? $_GET['search'] : '';
9 $category = isset($_GET['category']) ? $_GET['category'] : '';
10
11 // Build query with filters
12 $query = "SELECT p.*, c.name as category_name
13          FROM products p
14          LEFT JOIN categories c ON p.category_id = c.id
15          WHERE 1=1";
16
17 if ($search) {
18     $query .= " AND (p.name LIKE :search OR p.description LIKE :search)";
19 }
20 if ($category) {
21     $query .= " AND p.category_id = :category";
22 }
23
24 $query .= " ORDER BY p.created_at DESC";
25
26 $stmt = $db->prepare($query);
27
28 if ($search) {
29     $searchParam = "%$search%";
30     $stmt->bindParam(':search', $searchParam);
31 }
32 if ($category) {
33     $stmt->bindParam(':category', $category);
34 }

Tip: Try the Plan agent to research and plan before implementing changes.
+ index.php
Describe what to build
+ Auto
Local Default Approvals
unknown (4 hours ago) Ln 103, Col 138 Spaces: 4 UTF-8 CRLF PHP
Type here to search 16°C 11:31 04/06/2026
```

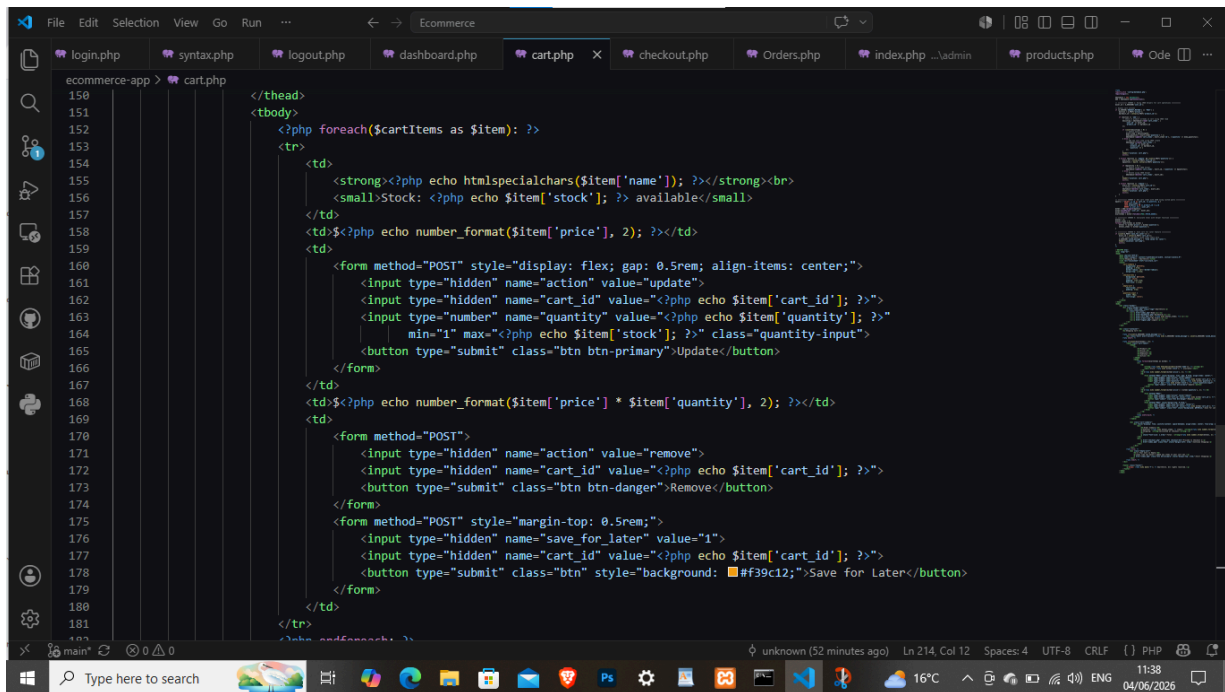
Folder structure screenshots



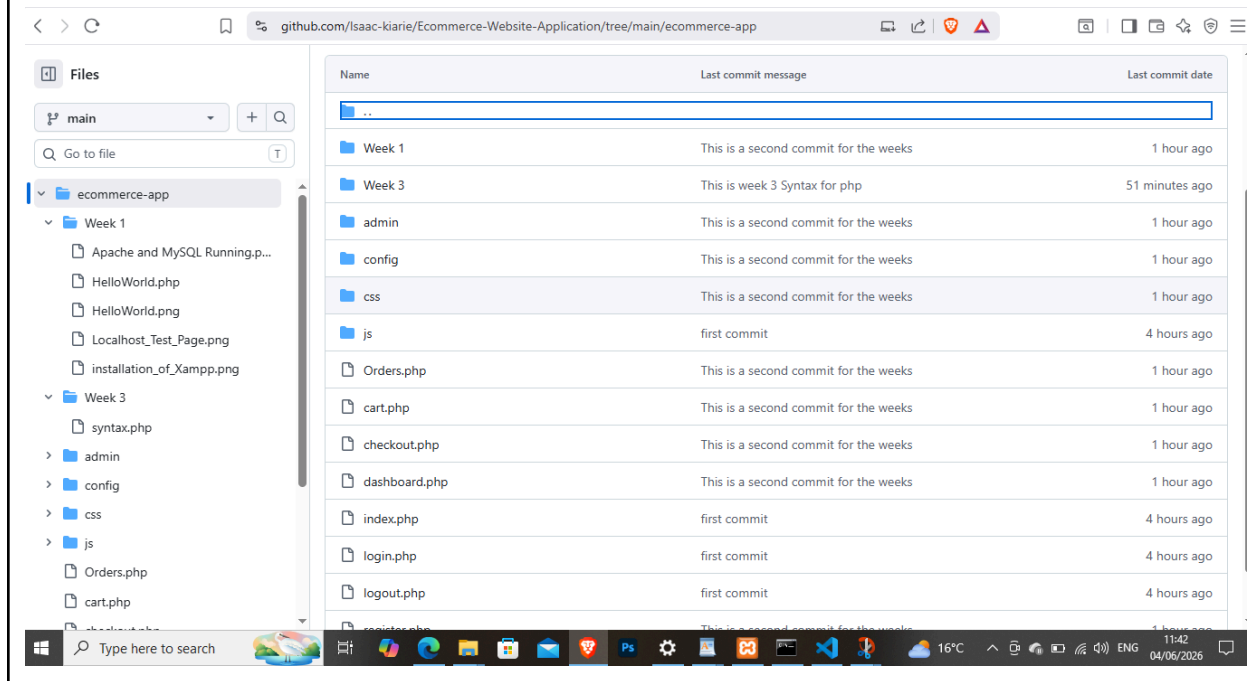
Database connection screenshots



Code snippets



GitHub updates



The screenshot shows a GitHub repository page for 'Isaac-kiarie/Ecommerce-Website-Application'. The left sidebar displays the file structure, including folders for 'Week 1', 'Week 3', 'admin', 'config', 'css', and 'js', along with various PHP and PNG files. The main area shows a table of commits with columns for 'Name', 'Last commit message', and 'Last commit date'.

Name	Last commit message	Last commit date
..		
Week 1	This is a second commit for the weeks	1 hour ago
Week 3	This is week 3 Syntax for php	51 minutes ago
admin	This is a second commit for the weeks	1 hour ago
config	This is a second commit for the weeks	1 hour ago
css	This is a second commit for the weeks	1 hour ago
js	first commit	4 hours ago
Orders.php	This is a second commit for the weeks	1 hour ago
cart.php	This is a second commit for the weeks	1 hour ago
checkout.php	This is a second commit for the weeks	1 hour ago
dashboard.php	This is a second commit for the weeks	1 hour ago
index.php	first commit	4 hours ago
login.php	first commit	4 hours ago
logout.php	first commit	4 hours ago

Student Reflection

.Within this week i got my hands dirty on the code making the Validation forms ensuring all the necessary information has been imputed by the user, I have understood about, backend processing, Process forms using PHP, request-response workflow, login logic, sessions basics, Structure projects professionally. I find it easier now with the concepts that used to seem complicated.

WEEK 5: User Authentication and Session Management

Database screenshots

The screenshot shows the phpMyAdmin interface for the 'ecommerce_db' database. The left sidebar lists the database structure, including tables like 'cart_items', 'categories', 'orders', 'order_items', 'products', and 'users'. The main panel displays the 'Structure' tab for the 'ecommerce_db' database. It shows a list of tables with their respective actions (Browse, Structure, Search, Insert, Empty, Drop) and a summary row for all 6 tables. The summary row indicates 12 rows, InnoDB engine, utf8mb4_general_ci collation, 208.0 KiB size, and 0 B overhead.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ cart_items	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ categories	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ order_items	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ products	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	Sum	12	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

Table Creation Shots.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ cart_items	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ categories	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ orders	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ order_items	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ products	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	Sum	12	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

XAMPP SCREENSHOT

XAMPP Control Panel v3.2.4 [Compiled: Jun 5th 2019]

XAMPP Control Panel v3.2.4

Modules

Service	Module	PID(s)	Port(s)	Actions
	Apache	2260 9868	80, 443	Stop Admin Config Logs
	MySQL	10784	3306	Stop Admin Config Logs
	FileZilla			Start Admin Config Logs
	Mercury			Start Admin Config Logs
	Tomcat			Start Admin Config Logs

Config Netstat Shell Explorer Services Help Quit

14:18:29 [main] All prerequisites found
14:18:29 [main] Initializing Modules
14:18:29 [main] Starting Check-Timer
14:18:29 [main] Control Panel Ready
14:35:49 [Apache] Attempting to start Apache app...
14:35:49 [Apache] Status change detected: running
14:35:50 [mysql] Attempting to start MySQL app...
14:35:50 [mysql] Status change detected: running

CRUD operation screenshots

The screenshot shows the 'SmartStore' application's shopping cart interface. The page has a dark blue header with navigation links: Home, Dashboard, Cart (2), My Orders, and Logout. The main content area is titled 'Shopping Cart' and features a table with the following columns: Product, Price, Quantity, Subtotal, and Action.

Product	Price	Quantity	Subtotal	Action
T-Shirt Stock: 100 available	\$19.99	1	\$19.99	Remove Save for Later
Novel Stock: 75 available	\$14.99	1	\$14.99	Remove Save for Later

Below the table is an 'Order Summary' section:

Order Summary
Subtotal (2 items): **\$34.98**
Shipping: Calculated at checkout
Total: **\$34.98**

At the bottom right of the summary, there are two buttons: 'Proceed to Checkout →' (green) and 'Continue Shopping' (purple).

SQL query screenshots

The screenshot shows the phpMyAdmin interface with the 'eCommerce_db' database selected. The 'SQL' tab is active, displaying a SQL query to create an 'orders' table. The query is as follows:

```
41 quantity INT DEFAULT 1,  
42 added_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
43 FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,  
44 FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE CASCADE,  
45 UNIQUE KEY unique_cart_item (user_id, product_id)  
46 );  
47  
48 -- Orders table  
49 CREATE TABLE orders (  
50 id INT PRIMARY KEY AUTO_INCREMENT,  
51 user_id INT,  
52 order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
53 total_amount DECIMAL(10,2),  
54 status ENUM('pending', 'processing', 'completed', 'cancelled') DEFAULT 'pending',  
55 shipping_address TEXT,  
56 )
```

Below the query editor, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters' and a text input for 'Bookmark this SQL query:'. At the bottom, there are checkboxes for 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks', along with a 'Go' button.

GitHub updates

The screenshot shows a web browser displaying the GitHub repository page for 'Isaac-kiarie/Ecommerce-Website-Application/tree/main/ecommerce-app'. The left sidebar shows the file structure under the 'main' branch, including folders for 'Week 1', 'Week 3', 'admin', 'config', 'css', and 'js', along with various PHP and PNG files. The main area displays a table of files and folders with their last commit messages and dates.

Name	Last commit message	Last commit date
..		
Week 1	This is a second commit for the weeks	2 hours ago
Week 3	This is week 3 Syntax for php	2 hours ago
admin	This is a second commit for the weeks	2 hours ago
config	This is a second commit for the weeks	2 hours ago
css	This is a second commit for the weeks	2 hours ago
js	first commit	6 hours ago
Orders.php	This is a second commit for the weeks	2 hours ago
cart.php	This is a second commit for the weeks	2 hours ago
checkout.php	This is a second commit for the weeks	2 hours ago
dashboard.php	This is a second commit for the weeks	2 hours ago
index.php	first commit	6 hours ago
login.php	first commit	6 hours ago
logout.php	first commit	6 hours ago

Student Reflection

Within this week i got chance to work on different operations such as Create, Update, Delete and read, i was unable to understand things like retrieving of data and information from the database towards the front of the website, it was exciting.

WEEK 6: Database Integration and CRUD Operations

Fig1 : Tables in the phmyadmin

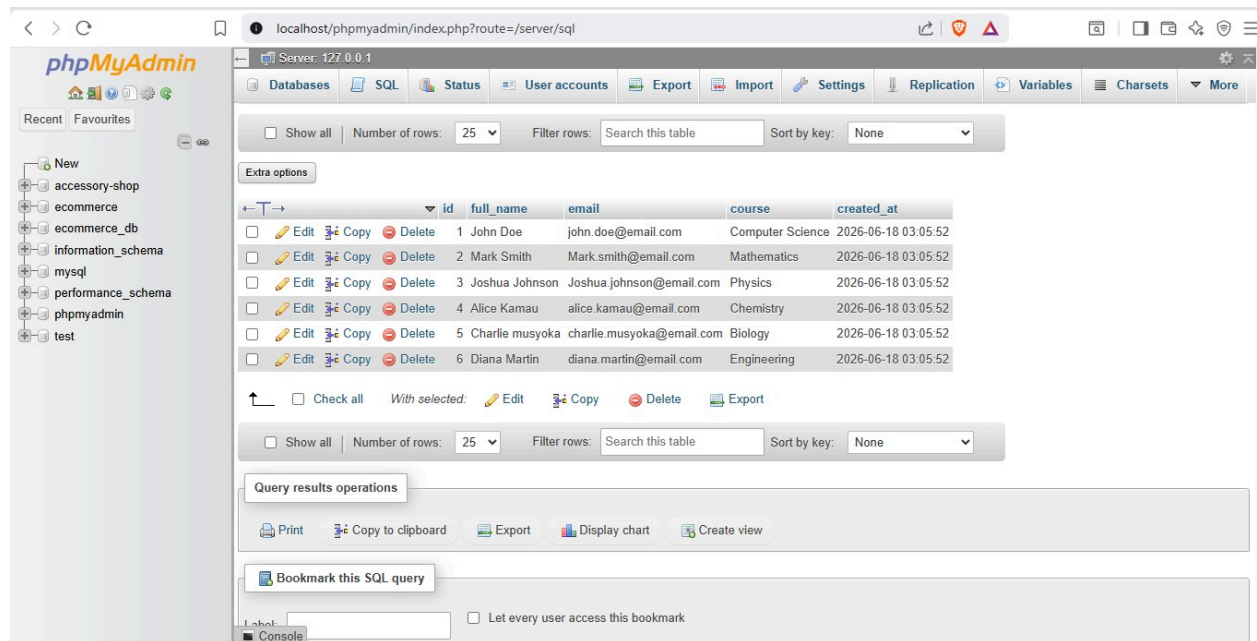


Fig 2: Operations to insert records.

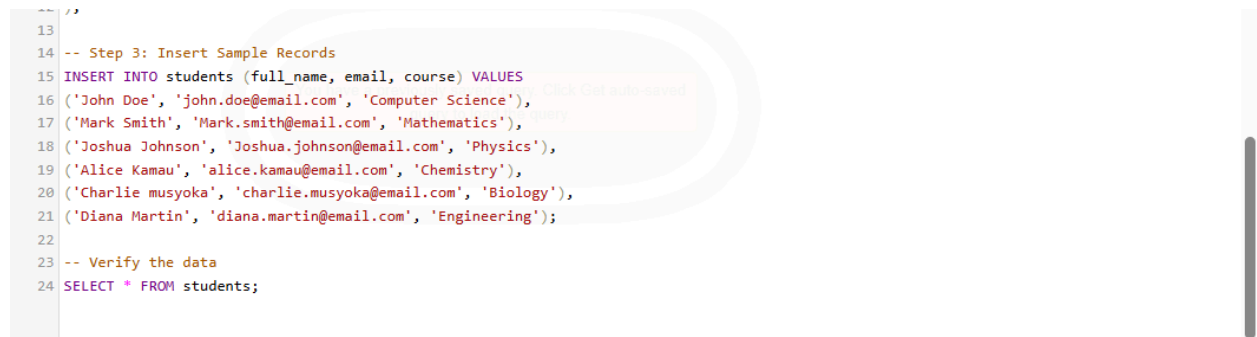
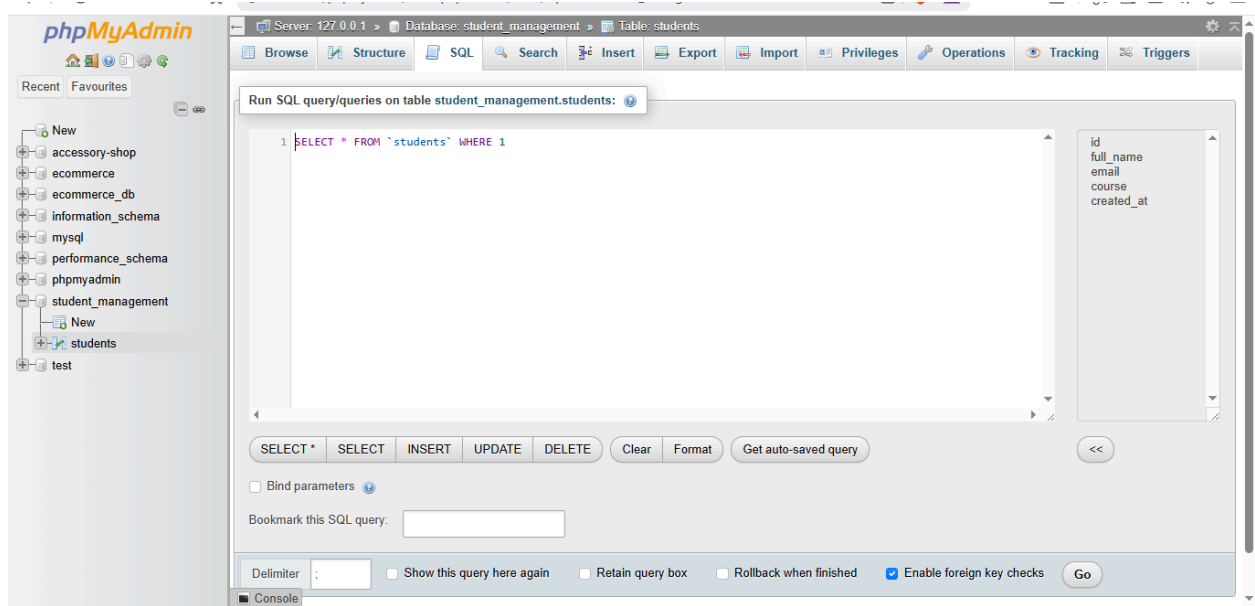


Fig 3: Read the operations records.



				id	full_name	email	course	created_at
☐	Edit	Copy	Delete	1	John Doe	john.doe@email.com	Computer Science	2026-06-18 03:05:52
☐	Edit	Copy	Delete	2	Mark Smith	Mark.smith@email.com	Mathematics	2026-06-18 03:05:52
☐	Edit	Copy	Delete	3	Joshua Johnson	Joshua.johnson@email.com	Physics	2026-06-18 03:05:52
☐	Edit	Copy	Delete	4	Alice Kamau	alice.kamau@email.com	Chemistry	2026-06-18 03:05:52
☐	Edit	Copy	Delete	5	Charlie musyoka	charlie.musyoka@email.com	Biology	2026-06-18 03:05:52
☐	Edit	Copy	Delete	6	Diana Martin	diana.martin@email.com	Engineering	2026-06-18 03:05:52
☐	Edit	Copy	Delete	7	isaac Kiarie	isaackiarie29@gmail.com	Business	2026-06-18 06:36:12

☐ Check all With selected: Edit Copy Delete Export

Fig 4: Update Operation

localhost/ECOMMERCE/student_management_system/edit_student.php?id=7

Demonstration 5 & Activity 3: Edit Student Record

✓ Student updated successfully!

Editing Student #7

Update the student information below.

Full Name *

isaac Kiarie

Email *

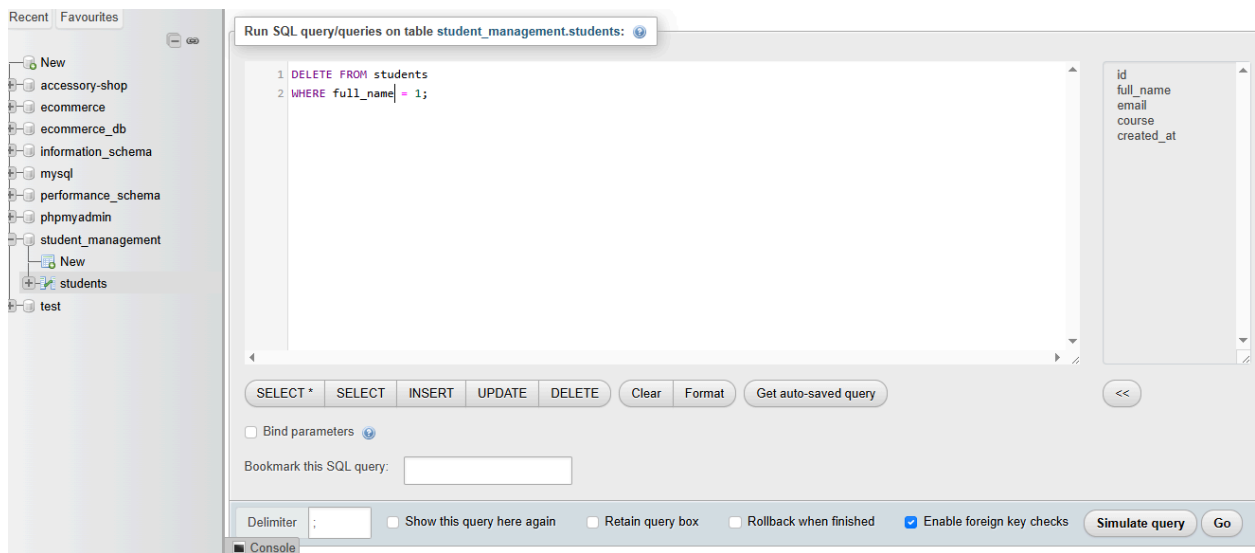
isaackiarie29@gmail.com

Course *

Computer Science

Update Student Cancel

Fig 5: Delete Operation.





Reflection:

In this week i got an opportunity to interact with different part of a fully developed system, i have seen the essence of having a well functioning database, learnt about the CRUD operation and how to connect the database with the web application, it was awesome.

WEEK 7: User Authentication and Session Management.

Fig 1: Registration Form

Register
Create your account to get started

Demonstration 1: User Registration with Password Hashing

Full Name *
Enter your full name

Email Address *
Enter your email

Password *
Minimum 8 characters
Must contain uppercase, lowercase, number, and special character

Confirm Password *
Confirm your password

Create Account

© 2026 Authentication System

Fig 2: Hashed Password.

	id	full_name	email	password	role	created_at	last_login
☐ Edit Copy Delete	1	isaac Kiarie	isaackiarie29@gmail.com	\$2y\$12\$ZZZPAoBKDxb3K0grdevWwOPPJKMGUfiW4dAy3jb0rS8...	student	2026-06-19 14:33:34	2026-06-19 14:33:41

↑
☐ Check all
With selected:
Edit
Copy
Delete
Export

Fig 3: Login

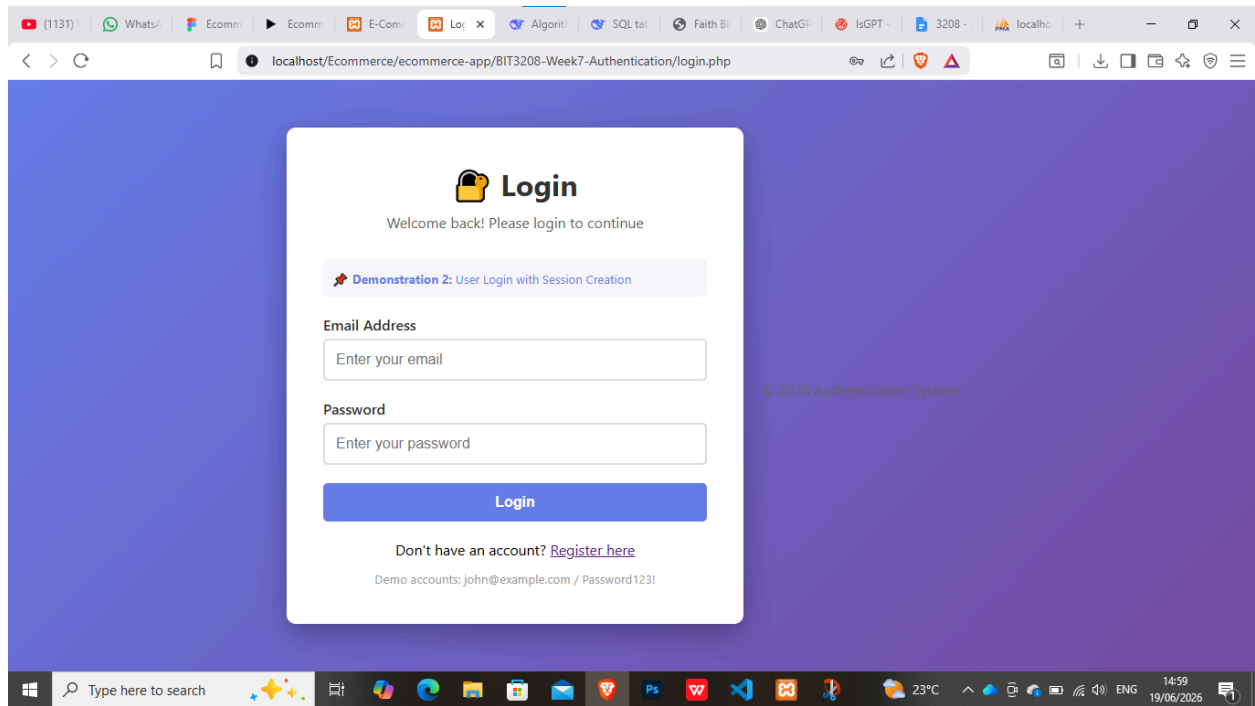


Fig 4: Dashboard.

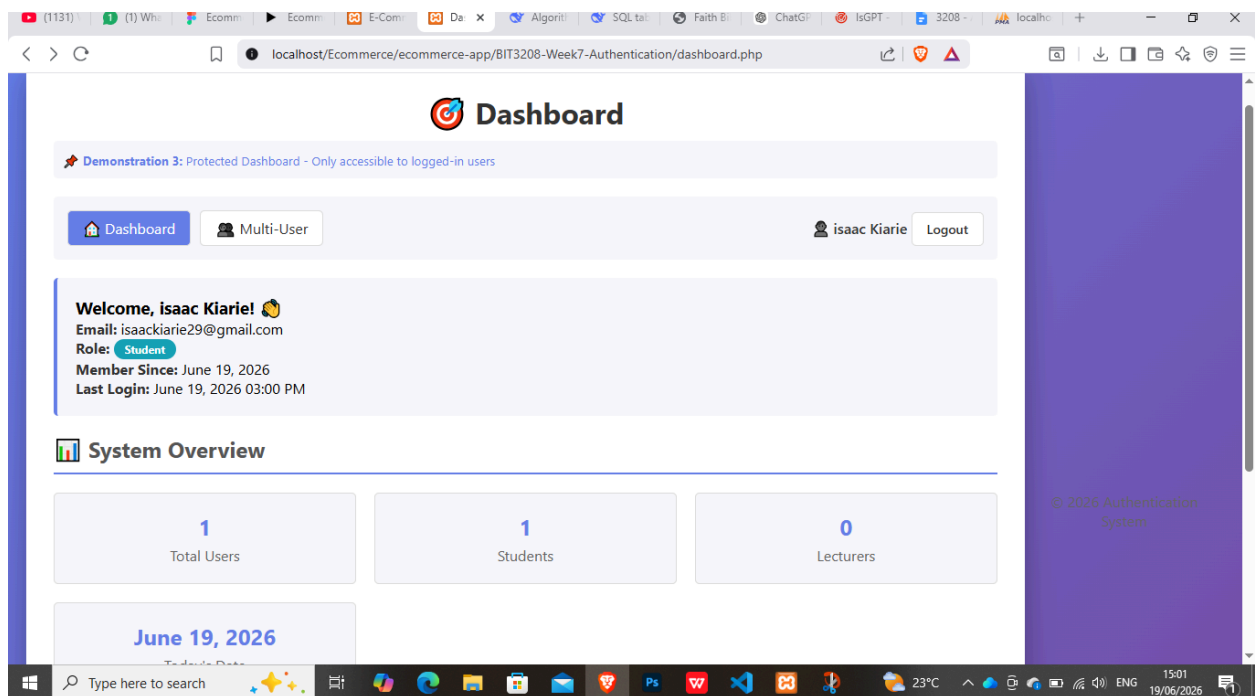
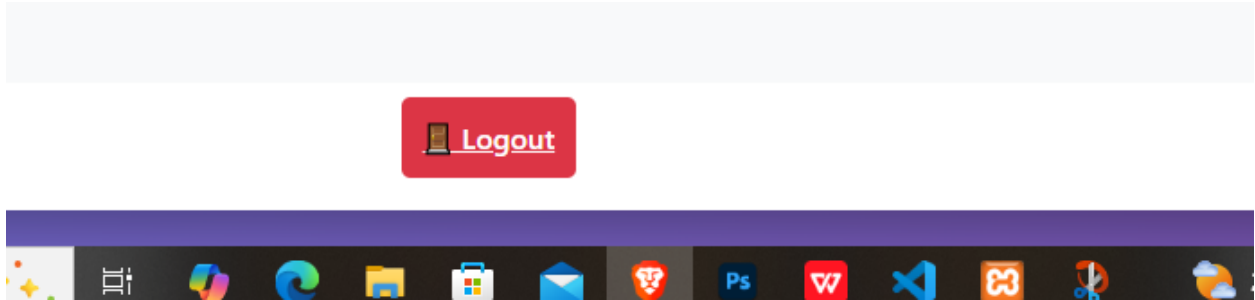


Fig 5: LogOut Operation.

**Reflection:**

I have learnt about authentication and authorization and how they differ in terms of their functionalities, Authentication it talks more of who are you? While authorization is concentrated of rights and permission to the right user who is validated and rightly allowed to access certain information within the system. I have also learnt about and how the passwords are store in a way that even when an sql injection is attempted the passwords are very secured.